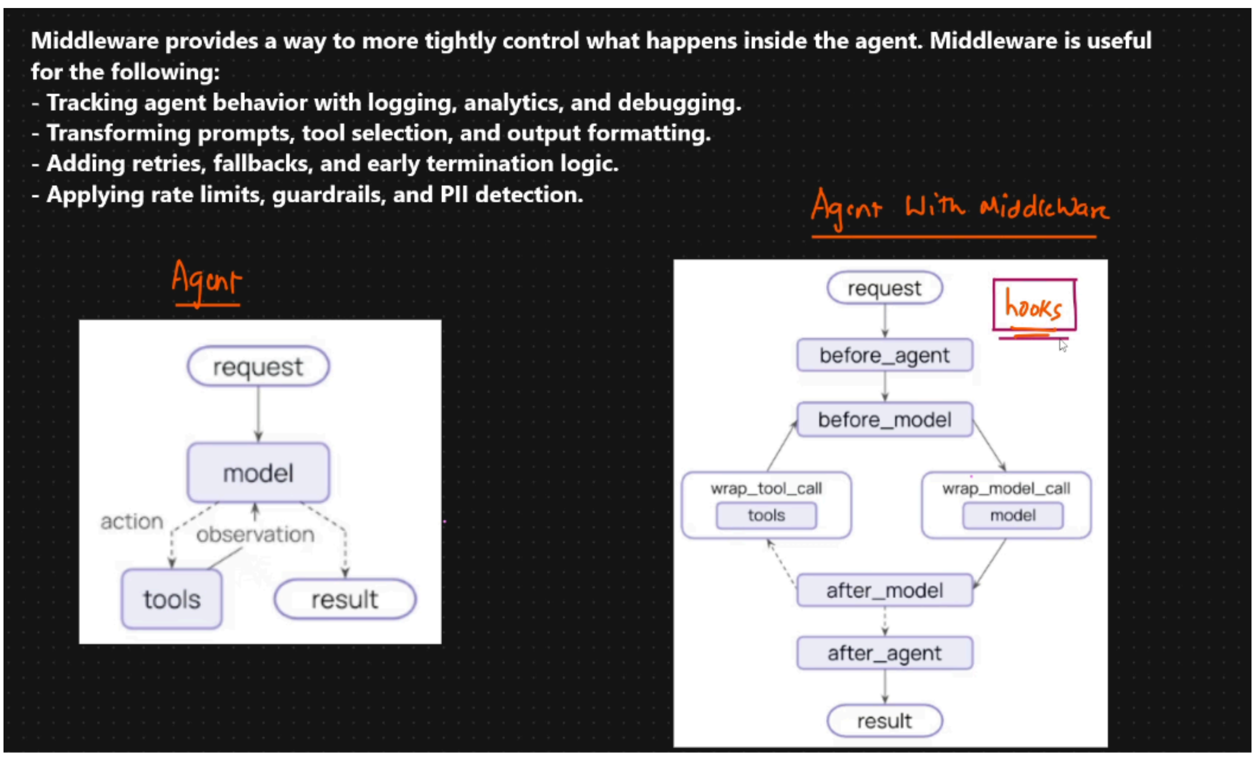
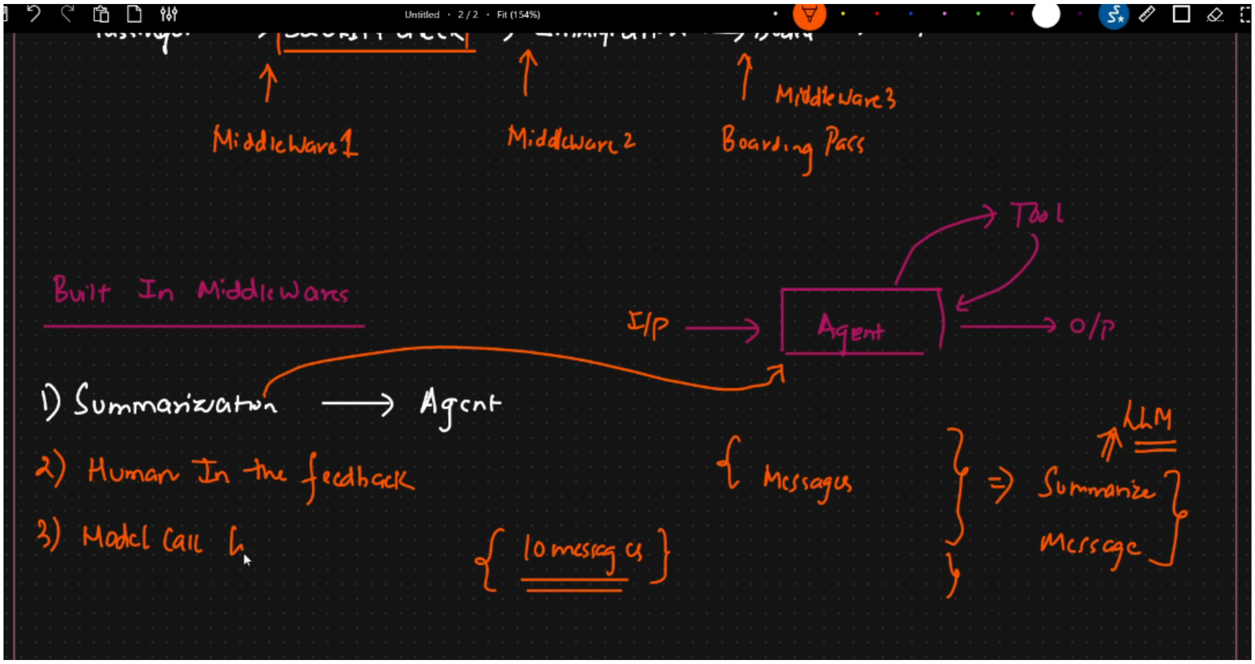
1. Download google antigravity IDE
2. install the package manager “UV”
3. “uv init”, this wil initialize a python project.
4. To create virtual environment, run command “uv venv”
5. to activate virtual environment “.venu\scripts\activate”
6. Create a(requirement.text) text file and the required libraries
7. to install all the libraries in the text file, “uv add -r requirements.text”
8. if you want to add any library individually, run “uv add library_name”
9. Now create 3 keys and add them .env file
    a. go to google ai studio and create a api key
    b. groq api key
    c. open api key
10. for Agent, LLM’s depend on 3rd party tool to get “context” for today’s news
11. before to create an agent they were using “react” architecture, and it was complex. These days they are using langchain version 1
12. Go through with each file in the project. It’s self explanatory.
13. summarization middleware using langchain





The following middleware work with any LLM provider:

Middleware	Description
Summarization	Automatically summarize conversation history when approaching token limits.
Human-in-the-loop	Pause execution for human approval of tool calls.
Model call limit	Limit the number of model calls to prevent excessive costs.
Tool call limit	Control tool execution by limiting call counts.
Model fallback	Automatically fallback to alternative models when primary fails.
PII detection	Detect and handle Personally Identifiable Information (PII).
To-do list	Equip agents with task planning and tracking capabilities.
LLM tool selector	Use an LLM to select relevant tools before calling main model.
Tool retry	Automatically retry failed tool calls with exponential backoff.
Model retry	Automatically retry failed model calls with exponential backoff.
LLM tool emulator	Emulate tool execution using an LLM for testing purposes.
Context editing	Manage conversation context by trimming or clearing tool uses.
Shell tool	Expose a persistent shell session to agents for command execution.
File search	Provide Glob and Grep search tools over filesystem files.

```python
from langchain.agents import create_agent
from langchain.agents.middleware import SummarizationMiddleware
from langgraph.checkpoint.memory import InMemorySaver
from langchain_core.messages import HumanMessage, SystemMessage

### Messagebased summarization
agent=create_agent(
    model="gpt-4o-mini",
    checkpointer=InMemorySaver(),
    middleware=[
        SummarizationMiddleware(
            model="gpt-4o-mini",
            trigger=("messages",10),
            keep=("messages",4)
        )
    ]
)
```